MULTIMODAL BIOSENSOR TELEMETRY

Smart acoustic audio and webcam facial distress tracking

Two continuous patient-portal pipelines convert spontaneous microphone audio and camera frames into quantified clinical biomarkers and real-time digital twin adaptation.

STREAM 04 • ACOUSTIC SENSING

Smart Acoustic Cough & Wheeze Sensor

Live microphone audio analyzed via Google YamNet (AudioSet Class 47/50) & Librosa DSP



- 512-bin STFT Spectrum: Real-time frequency power & transient energy cough burst detection
- Tracked Biomarkers: Hourly Cough Count (0 bursts), Graded Severity (0–3), Wheeze (400–1600 Hz)
- Prospective Risk Ingestion: Real-time 48h flare-up risk calculation (62.4% High Risk outlook)

STREAM 05 • FACIAL COMPUTER VISION

Smart Webcam Facial Distress Biosensor

Live webcam analysis tracking 468 3D landmarks via OpenCV/MediaPipe + optical rPPG



- 468 3D Landmark Mesh: Real-time 30 FPS facial geometry with IPD stabilization lock
- Tracked Biomarkers: Oral Breathing (MAR: 0.14 vs >0.35), Alar Flaring, SCM Accessory Strain
- Contactless Vitals: Forehead green-channel pulse (72 bpm) & nose-motion resp rate (16.0 brpm)

CLOSED-LOOP DIGITAL TWIN INTEGRATION

Acoustic cough bursts and facial respiratory distress telemetry stream into the 12-hour rolling wearable feature window, modulate the latent mucosal inflammation state I(t) in the Euler-Maruyama ODE, and trigger clinician outlier retraining via Elastic Weight Consolidation (EWC).